

A hydrogen supply system utilizing PEMFC exhaust heat and modular metal hydride tanks for hydrogen-powered bicycles[☆]

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HIGHLIGHTS

- A PEMFC–MH hydrogen supply system designed for hydrogen-powered bicycles (H-bike).
- Exhaust heat of PEMFC is recovered via split-tank and honeycomb MH configurations.
- The split-tank strategy reduces thermal/pressure gradients, thereby improving desorption uniformity.
- A honeycomb arrangement enhances exhaust heat utilization and improves system energy efficiency.
- A two-layer MH design achieves a 39.90 % thermal utilization efficiency at a 215 W PEMFC output.

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ABSTRACT

A compact hydrogen supply system for thermally integrating metal hydride (MH) tanks with a proton exchange membrane fuel cell (PEMFC) for a hydrogen-powered electric-assist bicycle (H-bike) is proposed. The system recovers the exhaust heat generated by the PEMFC to sustain hydrogen desorption and improve the system's energy efficiency. The results demonstrate that the split-tank strategy decreases thermal and pressure gradients and enhances heat transfer and hydrogen release. The honeycomb tank configuration further improves hydrogen desorption by promoting uniform airflow distribution around each tank, thereby improving exhaust heat utilization from the PEMFC. It employs a layer-adjustable configuration, facilitating the flexible adaptation of MH cartridge quantities to meet hydrogen demand and prevailing road conditions in urban areas. Under a PEMFC power output of 215 W, the system maintains a stable hydrogen flow rate for over 30 min, with a heat recovery efficiency of 22.62 %. Furthermore, increasing the number of MH cartridge layers significantly improves the thermal utilization of the system, achieving a utilization efficiency of 39.90 % with two layers. These findings confirm the feasibility and scalability of the proposed system for H-bike, highlighting its potential as a decentralized hydrogen supply solution for lightweight mobility and urban transportation applications.

1. Introduction

With the growing demand for sustainable transportation and zero-carbon urban mobility in smart cities, the development of compact and lightweight energy systems has become paramount for small-scale mobile platforms, including drones, scooters, and electric-assist bicycles [1–3]. Despite widespread lithium-ion battery use, they often face

energy density and refueling time limitations, especially during prolonged use [4,5]. Hydrogen energy offers a high energy density, quick refueling, and zero emissions, making it ideal for lightweight mobility [6–8]. In recent years, hydrogen-powered electric-assist bicycles (H-bikes) have garnered considerable attention as a viable solution combining clean energy with practical functionality [9,10]. However, successful implementation depends on two key technologies: proton

Abbreviations: FC, Fuel cell; MH, Metal hydride; PEMFC, Proton Exchange Membrane Fuel Cell; PCM, Phase change materials; H-bike, hydrogen-powered electric-assist bicycle; MHT, Metal Hydride Tank; HX, Heat Exchanger; SOFC, Solid Oxide Fuel Cell; ORC, Organic Rankine Cycle; TEG, Thermoelectric Generator; RSOFC, Reversible Solid Oxide Fuel Cell; PCT, Pressure Composition Temperature.

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exchange membrane fuel cells (PEMFCs) for hydrogen-to-electricity conversion and metal hydride (MH) alloys for safe, reversible storage [11–13].

PEMFCs are crucial in hydrogen energy due to their high efficiency, low operating temperatures, compactness, and rapid startup, making them ideal for transport and portable applications [14–16]. Despite their electrochemical benefits, PEMFCs produce substantial heat, emitting about 40 % to 60 % of the hydrogen energy as exhaust heat [17–19]. Ineffective heat management can cause local overheating, membrane dehydration, catalyst degradation, and reduce system lifespan. Not using dissipated heat also lowers energy efficiency [20–22]. In conventional designs, thermal energy is usually removed by air or liquid and discharged into the surrounding environment, representing a lost opportunity for energy recovery [23,24]. Various methods have been proposed for heat recovery; however, many are limited by their reliance on complex or large system setups [25–27]. Therefore, creating passive, compact, and integrated heat recovery solutions is essential for small-scale PEMFCs.

MH alloys represent another key technology and have attracted substantial attention as a promising medium for hydrogen storage, particularly in small-scale and portable applications [28,29]. Their main advantages are high hydrogen density, safe low-pressure operation, and good reversibility over cycles [30,31]. However, their inherent thermophysical limitations greatly limit practical implementation. One of the key challenges is the endothermic hydrogen desorption, where insufficient external heat causes the MH bed temperature to drop, lowering equilibrium pressure and hydrogen release rate [32,33]. In the absence of external heating, their desorption rates may be insufficient to meet the real-time hydrogen demand of PEMFCs, particularly under dynamic or peak operating conditions [34,35]. Another key challenge is the inherently low thermal conductivity of most MH materials [36,37]. This poor thermal conductivity results in significant internal temperature gradients during hydrogen desorption. Internal temperature gradients create spatial variations in local equilibrium pressure during desorption, which reduces the thermodynamic driving force [38]. As a result, the core region of the MH bed cools much faster than the outer shell, leading to uneven thermal distribution, nonuniform hydrogen desorption, and ultimately, reduced hydrogen utilization efficiency [39,40]. A nonuniform pressure distribution within MH tanks also poses a key challenge [41]. The geometry of the MH tank contributes to pressure non-uniformity. As hydrogen diffuses from the MH bed toward the outlet, a pressure gradient is established due to the rapid discharge of hydrogen at the outlet. Such pressure disparity negatively impacts desorption efficiency [42,43]. This problem is further exacerbated in elongated or large tanks, where the increased diffusion resistance impedes hydrogen movement, resulting in pressure distribution and reduced hydrogen desorption [44,45]. Addressing these combined thermal and pressure limitations is crucial for advancing MH-based hydrogen storage systems.

Numerous strategies have been proposed to improve thermal management, including adding internal fins with liquid cooling, designing multi-tubular porous MH reactors with optimized thermophysical properties, using activated-carbon reactors combined with PCM and metal fins, and developing hybrid PCM-metal foam composites to enhance heat transfer [46–49]. However, these methods involve complex reactor designs or complicated fabrication processes, which limit their use in compact and mobile systems. In contrast, combining PEMFC-generated exhaust heat with hydrogen desorption in MH tanks offers a simpler and more practical way to improve overall system efficiency, especially for mobile applications. Progress has been made in several studies by utilizing thermal management techniques, resulting in improved heat transfer and energy efficiency in hydrogen storage systems. For direct heat exchange, Liu et al. proposed a heat-coupled PEMFC stack integrated with embedded MH containers, facilitating direct thermal transfer. While improving heat utilization during hydrogen desorption, this embedded setup faces challenges related to

hydrogen absorption and the quick replacement of MH containers, which limit flexibility and maintainability. Additionally, the higher temperatures during hydrogen storage may negatively impact the durability of the PEMFC stack, decreasing its service life [50]. For heat pipe systems, Wang et al. employed heat pipes to recover exhaust heat during hydrogen desorption. The inherently passive nature of heat pipes restricts their controllability in real-time thermal management. Furthermore, heat pipes occupy a volume that could otherwise be utilized for MH alloys, and their incorporation diminishes the overall hydrogen storage efficiency [51]. For liquid heat transfer, Wang et al. implemented a water-circulation PEMFC-MHT system, facilitating controlled exhaust heat recovery and hydrogen desorption. Nonetheless, this system requires components such as pumps, valves, radiators, and piping, increasing cost requirements and power consumption [52]. For air-based heat transfer, Giap et al. demonstrated the thermal coupling between a reversible solid oxide fuel cell (RSOFC) and a LaNi₅-based MH system, utilizing exhaust steam to facilitate hydrogen desorption. However, the inherent complexity of RSOFC architectures renders them unsuitable for mobile or low-power applications [53]. Omrani et al. validated the feasibility of thermal coupling between a 2-kW open-cathode PEMFC and LaNi₅ canisters using a direct hot air flow. However, their study overlooked the impact of tank sizes and arrangements, leading to significant temperature gradients within the canisters; such thermal inconsistencies may negatively affect hydrogen desorption rates [54]. While these studies have provided valuable insights into the systems, they do not adequately address the challenges arising from concurrent PEMFC and MH components.

Considering the above, this study presents a compact, thermally integrated PEMFC-MH system that aligns with the practical requirements of H-bikes, utilizing the exhaust heat generated by the PEMFC to facilitate hydrogen desorption from the MH tanks. A split-tank strategy was proposed to address the inherent challenges of MH materials, coupled with a honeycomb-style configuration to improve the utilization of PEMFC exhaust heat and enhance the system's overall energy efficiency. Multiple experimental scenarios, including variations in hydrogen flow rate, pressure, temperature distribution, and the impact of PEMFC output power on exhaust heat availability, are evaluated to validate the feasibility of the proposed design.

2. Methodology

2.1. Split-tank strategy for the MH tank

In natural and forced convection, as well as turbulent flow, the exposed surface area significantly influences heat transfer efficiency between the tank and its surrounding environment. An increased surface area enhances the effectiveness of convective heat exchange, which is particularly crucial for facilitating the endothermic hydrogen desorption process in MH tanks [55]. Fig. 1 illustrates the geometry of the MH tanks and the placement of their internal thermocouples. A traditional single large MH tank was subdivided into six smaller tanks, maintaining an equivalent total volume to enhance thermal exchange during hydrogen desorption. This modular redesign involves an approximately 3.5-fold increase in the total surface area compared to the traditional large tank. The depicted dimensions represent the inner diameters, as the thermal conductivity of the aluminum shell significantly exceeds that of the MH alloy, rendering the internal structure a more critical factor for thermal conductivity. In addition to the increased surface area, the smaller tanks offer several advantages in terms of heat transfer characteristics. Firstly, the reduced diameter and height of each tank considerably shorten the heat conduction pathway from the tank wall to its center, promoting quicker heat transfer and a more uniform temperature distribution throughout the MH tank, along with an expected improvement in the desorption capacity. Secondly, the compact geometry of the small tanks facilitates a more uniform pressure distribution and hydrogen flow, preventing the uneven utilization frequently

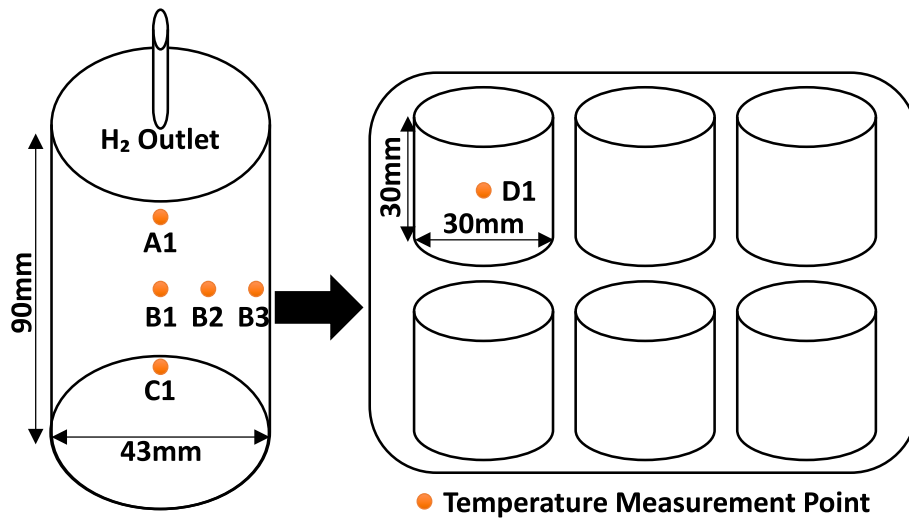


Fig. 1. Geometry of the MH tanks and their thermocouple positions.

observed in the elongated ends of large cylindrical tanks [44,45]. Thirdly, while a conventional cylindrical tank mainly exposes its lateral surface to the surrounding air, the spatial design of the smaller tanks allows nearly all sides to face the external environment. This greatly increases the pathways for convective heat transfer, enhancing the system's dynamic responsiveness to environmental changes. These thermal enhancements are expected to improve the temperature stability and hydrogen release of the MH tanks. Furthermore, this design is anticipated to facilitate the integration of exhaust heat from the PEMFC.

All MH tanks were filled with LaNi_5 , a substance widely employed in small-scale mobile applications due to its high volumetric hydrogen density and its capacity for efficient hydrogen absorption and desorption within a near-ambient temperature range [28,56]. Thermocouples were strategically incorporated at specific locations within the tanks to assess their internal thermal characteristics. Further, a comprehensive temperature mapping strategy was implemented in the larger tank to capture the axial and radial temperature distributions. Three sensors (A1, B1, and C1) were installed along the longitudinal axis at distances of 15, 45, and 75 mm from the hydrogen outlet, respectively. Furthermore, two sensors (B2 and B3) were embedded along the radial axis at depths of 11 and 1 mm, respectively, from the inner tank wall. This configuration facilitated the monitoring of the internal thermal conditions along both axes. In the small tank configuration, a central thermocouple (D1) was embedded at the geometric center to monitor the core temperature during hydrogen desorption.

2.2. Honeycomb tank configuration and PEMFC exhaust integration

The PEMFC in the system is rated at 400 W, serving as the primary power source for the H-bike and representing the leading thermal energy provider for hydrogen desorption. A power of 400 W is sufficient to meet the propulsion needs of the H-bike on various terrains while supporting the integration of auxiliary components, such as cooling fans [57]. The PEMFC uses an air-cooling mechanism with dual fans to continuously discharge exhaust gas that carries heat energy. A semi-enclosed ducted heat recovery system was used to ensure that the exhaust gas is directed through the duct containing the MH tanks, thereby maximizing the utilization of the PEMFC exhaust heat. Maintaining the stable operation of the PEMFC is crucial. Excessively high temperatures within the FC stack may have negative impacts, including reduced electrochemical efficiency; unstable water management; and the accelerated degradation of key materials, such as membranes and catalysts [58–60]. High temperatures can also disrupt the internal pressure balance and increase ohmic losses, reducing the energy conversion efficiency [61,62].

Therefore, designing a heat utilization strategy to both improve the heat recovery rate of the PEMFC exhaust and ensure its cooling performance is crucial.

This goal was achieved by adopting a honeycomb configuration; as shown in Fig. 2, four small MH tanks, each with an outer diameter of 35 mm, form a layer and are arranged symmetrically in a cylindrical duct with a diameter of 100 mm. A uniform gap of 4.5 mm is maintained between each pair of adjacent tanks as well as between each tank and the duct wall. This symmetrical and compact arrangement ensures an even distribution of airflow around all tanks. Notably, the total surface-to-volume ratio of the small tanks is significantly higher than that of a single large tank with the same total capacity. The honeycomb structure enhances interfacial heat transfer, stimulating local airflow turbulence and further improving heat transfer. This configuration enables the MH tanks to absorb heat from the PEMFC exhaust without leading to heat accumulation, a scenario that could otherwise affect the cooling behavior of the PEMFC. Two ducts were installed in parallel, each capable of accommodating up to five layers of MH tanks (with four tanks per layer), yielding 40 tanks in the current configuration. The number of layers can be flexibly adjusted based on hydrogen demand and available thermal input, providing adaptability for different operating scenarios. This approach is designed to facilitate stable hydrogen desorption and compact system integration, which are advantageous characteristics for lightweight mobility applications, including H-bikes.

2.3. Calculation of exhaust heat and absorbed heat

A PEMFC stack typically operates at a temperature exceeding that of the ambient environment, releasing thermal energy to the surroundings from its external surfaces via both convective and radiative mechanisms. A considerable proportion of the generated heat is employed for the internal phase change process associated with water evaporation, which predominantly occurs within the membrane and catalyst layers of the PEMFC [63,64]. The residual heat, unabsorbed by the internal components, is expelled through forced airflow as exhaust, consequently aiding in regulating and maintaining the stack temperature within its optimal operational limits [65]. This recoverable exhaust heat can be quantitatively assessed using the formulas in Eqs. 1 and 2 [56]. The symbols, definitions, and units used in the equations are summarized in Table 1.

$$\dot{Q}_{\text{out}} = \dot{m}_{\text{air}} C_{p,\text{air}} (T_{\text{out}} - T_{\text{in}}) \quad (1)$$

$$\dot{m}_{\text{air}} = \rho_{\text{air}} \bullet A \bullet v_{\text{air}}, \quad (2)$$

Hydrogen desorption from MH beds is a heat-driven process that

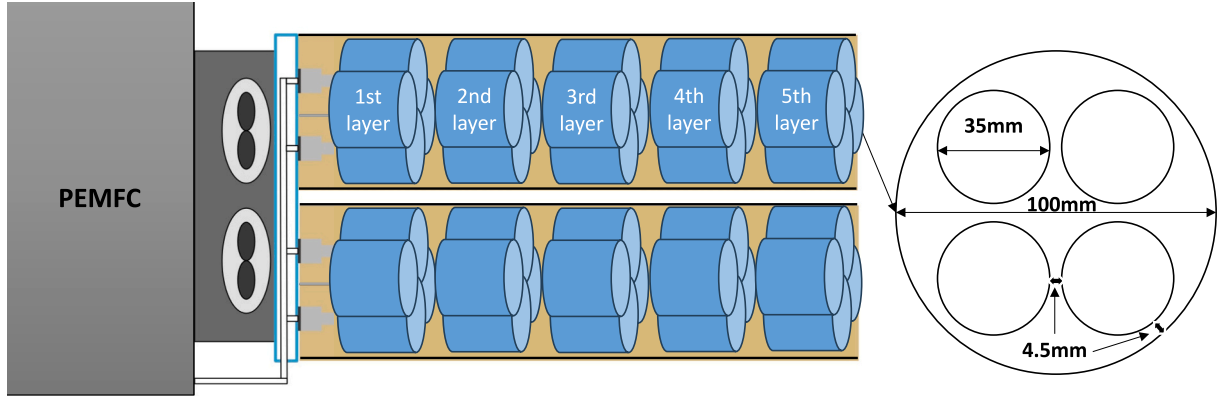


Fig. 2. Honeycomb configuration used in the PEMFC-MH system.

Table 1

Symbols, definitions, and units used in the governing equations [52,53,66].

Symbol	Definition	Unit	Value
$\dot{Q}_{out}$	Thermal power carried away by PEMFC exhaust air	W	–
$\dot{m}_{air}$	Mass flow rate of exhaust air	kg·s ⁻¹	–
$C_{p,air}$	Specific heat capacity of air at constant pressure	J·kg ⁻¹ ·K ⁻¹	1005
ρ_{air}	Air density	kg·m ⁻³	1.184
A	Effective cross-sectional area of exhaust outlet	m ²	0.0038
v_{air}	Exhaust air velocity	m·s ⁻¹	–
T_{in}, T_{out}	Inlet and outlet air temperatures	K	–
ϕ	Porosity of the MH bed	–	0.5
ρ_{H_2}	Hydrogen gas density	kg·m ⁻³	0.0838
$\vec{v}$	Superficial (Darcy) velocity vector	m·s ⁻¹	–
S_m	Mass source term of desorbed hydrogen	kg·m ⁻³ ·s ⁻¹	–
$\dot{n}_{H_2,out}$	Outlet hydrogen molar flow rate	mol·s ⁻¹	–
V	Total MH bed volume	m ³	–
ρ_{MH}	Density of MH solid	kg·m ⁻³	8300
wt	hydrogen storage capacity by weight	%	–
M_{H_2}	Molar mass of hydrogen	kg·mol ⁻¹	2.016 × 10 ⁻³
F	Hydrogen mass fraction in the MH	–	–
P	Gas pressure in pores	Pa	–
R	Universal gas constant	J·mol ⁻¹ ·K ⁻¹	8.314
K	Bed permeability	m ²	–
μ	Dynamic viscosity of hydrogen gas	Pa·s	–
$(\rho C_p)_{eff}$	Effective volumetric heat capacity of the MH bed	J·m ⁻³ ·K ⁻¹	–
λ_{eff}	Effective thermal conductivity of the MH bed	W·m ⁻¹ ·K ⁻¹	–
C_{p,H_2}	Specific heat capacity of hydrogen gas	J·kg ⁻¹ ·K ⁻¹	14,300
ΔH_{MH}	Desorption enthalpy of MH	J·mol ⁻¹	30.8 × 10 ³
$\dot{Q}_{total}$	Total thermal power absorbed by MH cartridge	W	–

involves mass transfer, gas flow, and heat transport within the porous medium. The governing equations of continuity, momentum, and energy conservation are summarized below, along with the definition used for calculating hydrogen concentration [66].

Continuity equation:

$$\phi \frac{\partial \rho_{H_2}}{\partial t} + \nabla \cdot (\phi \rho_{H_2} \vec{v}) = S_m \quad (3)$$

Overall hydrogen balance:

$$\dot{n}_{H_2,out} = (1 - \phi) V \frac{\rho_{MH} \cdot wt}{M_{H_2}} \left(-\frac{dF}{dt} \right) + \frac{d}{dt} \left(\frac{\phi VP}{RT} \right) \quad (4)$$

Momentum balance (Darcy's law):

$$\vec{v} = -\frac{K}{\mu} \nabla P \quad (5)$$

Equation of state:

$$\rho_{H_2} = \frac{M_{H_2} P}{RT} \quad (6)$$

Equation of energy conservation:

$$(\rho C_p)_{eff} \frac{\partial T}{\partial t} + \rho_{H_2} C_{p,H_2} \frac{1}{\phi} \vec{v} \cdot \nabla T = \nabla \cdot (\lambda_{eff} \nabla T) - \frac{\Delta H_{MH} S_m}{M_{H_2}} \quad (7)$$

The total heat absorbed by the cartridge:

$$\dot{Q}_{total} = \frac{\Delta H_{MH}}{M_{H_2}} \int_V S_m dV + (\rho C_p)_{eff} V \frac{dT}{dt} \quad (8)$$

Since the sensible heat contribution of the solid matrix is relatively small and the temperature distribution inside the cartridge is non-uniform, this term was omitted in the subsequent calculations. If it were to be considered, the total absorbed heat would increase when the bed temperature rises above its initial value and decrease when the bed temperature falls below the initial value.

2.4. Experimental apparatus and experimental procedures

The experimental platform utilized in this study is schematically depicted in Fig. 3. It comprises a PEMFC power module, modular MH tanks, a power load controller, a hydrogen flow regulation unit, and a temperature and pressure monitoring system. The PEMFC is the electrical power supply and the primary thermal energy source for hydrogen desorption. This PEMFC is air-cooled via dual integrated fans, which provide oxygen for the electrochemical reaction. MH tanks are arranged in a honeycomb-style configuration and aligned within a semi-enclosed exhaust duct to ensure optimal heat recovery. Two such ducts were installed in parallel, each supporting up to five layers of MH cartridges. A pressure-reducing valve was installed in the hydrogen supply line upstream of the PEMFC inlet to ensure that hydrogen is delivered at the appropriate pressure. Flow meters and pressure gauges were positioned between the tanks and the fuel cell to continuously monitor the hydrogen flow rate and internal pressure. Furthermore, thermocouples were embedded near the center of each MH tank and at the cartridge joints to capture real-time temperature data. All signals from thermocouples, pressure transducers, and flow sensors were collected through a data acquisition system interfaced with LabVIEW software for real-time monitoring and recording.

A decoupled hydrogen flow configuration was employed to evaluate the hydrogen desorption behavior of the MH tanks under various conditions. Before testing, the MH tanks were fully charged with hydrogen at 0.3 MPa. The PEMFC received hydrogen from a high-pressure hydrogen cylinder during the desorption experiments, while the MH tanks were positioned in the exhaust flow duct. The hydrogen released from the MH tanks was discharged into the atmosphere through a

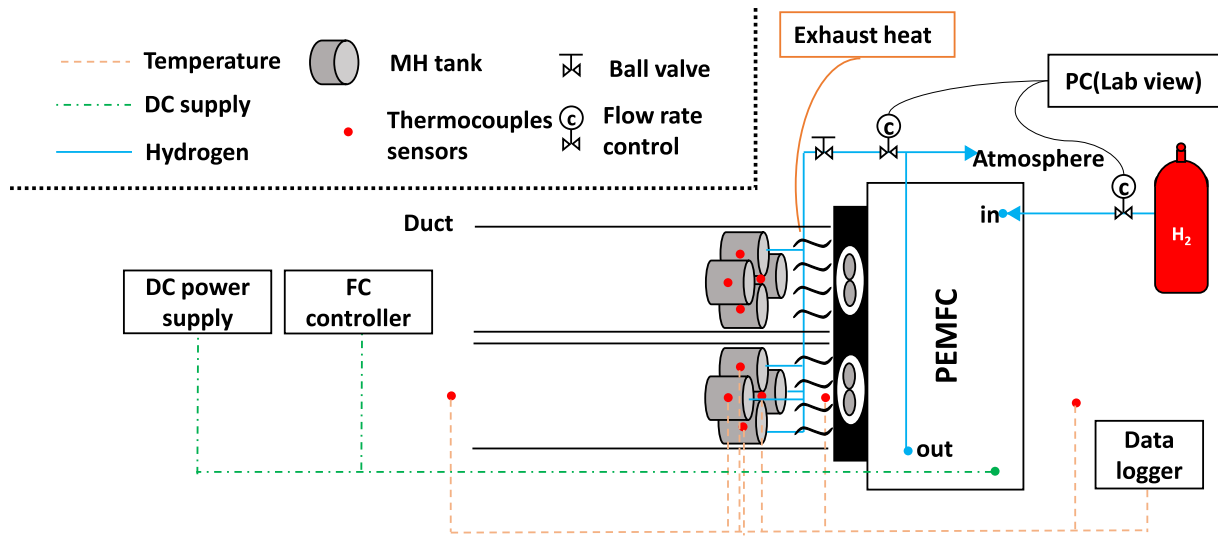


Fig. 3. Experimental apparatus.

calibrated mass flow controller. This approach effectively eliminated the fluctuations in hydrogen demand that would occur if the PEMFC were directly supplied by the MH tanks, thereby ensuring a stable and controllable power output. By decoupling the hydrogen supply, the approach allowed for a more precise evaluation of the MH tanks' ability to sustain hydrogen release under varying PEMFC power conditions.

3. Results and discussion

3.1. Hydrogen desorption behavior and superiority of the split-tank strategy

Hydrogen desorption experiments were conducted on both the large and small tank configurations to assess the effectiveness of the split-tank strategy. The tests were executed under natural convection at room temperature, with an initial internal hydrogen pressure of 0.3 MPa. Hydrogen was released into the atmosphere at maximum flow rates of 1000 and 1125 sccm for the large and small tanks, respectively. The temperature was monitored in real time using thermocouples placed at predefined positions within the tanks, as illustrated in Fig. 1. Furthermore, natural convection and room temperature conditions were chosen to eliminate uncertainties arising from forced or turbulent airflow interactions with various tank geometries. This approach ensured that performance differences were due to the tanks' geometric and thermal features, providing a reliable basis for predicting behavior in complex conditions like forced convection or turbulent flow.

Fig. 4(a) illustrates the radial temperature distribution within the

large tank, monitored at three distinct radial positions: B1, B2, and B3. All three locations exhibit a linear decrease in temperature at the start of hydrogen desorption, during which a steady 1000 sccm flow rate is maintained. The endothermic desorption continuously absorbs heat, causing the MH bed temperature to decrease. The different slopes of the curves indicate spatial variations: the center (B1) cools more rapidly, while the wall-adjacent point (B3) cools more slowly. This behavior results from limited thermal conductivity in the core and heat absorption from ambient air near the wall. Around 200 s, the temperature profiles level off, indicating a quasi-steady state with reduced hydrogen flow as the cooling MH lowers the equilibrium pressure. The center point B1 stabilizes about 4 °C below B2 and B3, further suppressing hydrogen desorption in this region. The radial temperature gradient confirms the presence of pronounced thermal nonuniformity in large MH tanks under natural convection. As a result, only material near the walls remains warmer for desorption, while the cooler core remains inactive, thereby reducing the overall hydrogen release capacity.

The axial temperature distribution within the large tank was assessed, as illustrated in Fig. 4(b). Temperatures were monitored at three axial positions: A1, B1, and C1. All showed an initial linear decline, but cooling was fastest at A1 near the outlet and slowest at C1. This results from an axial pressure gradient during desorption: rapid pressure drop at A1 speeds up hydrogen release and causes endothermic cooling. Meanwhile, regions farther away retain higher pressures, which limit desorption and keep temperatures higher. Gas transport resistance along the extended flow path further worsens this imbalance [44,45]. Around 200 s, the temperature curves plateau, indicating a quasi-steady state

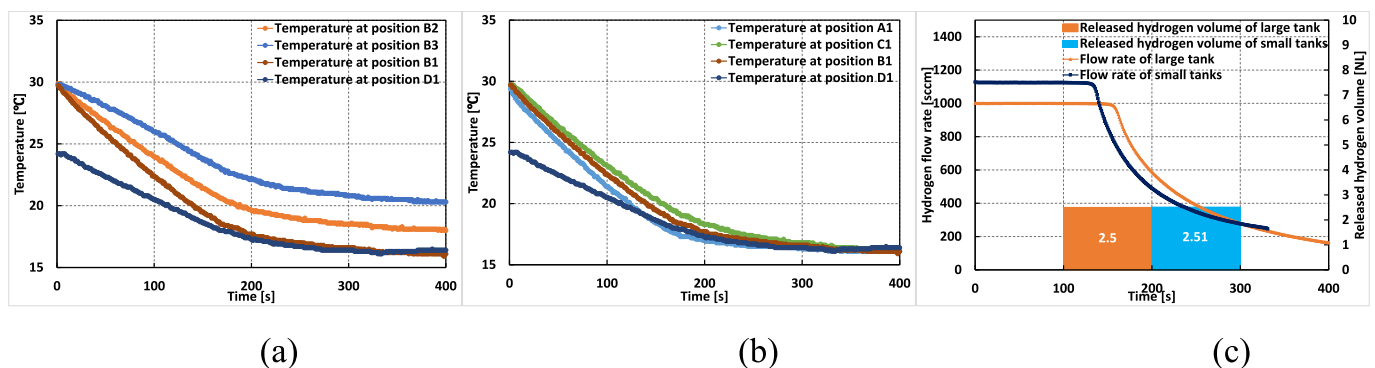


Fig. 4. Thermal and desorption performance of large and small MH tanks under natural convection: (a) radial temperature variation, (b) axial temperature variation, and (c) hydrogen flow rate and cumulative volume of released hydrogen.

consistent with the trend observed in Fig. 4(a). These results highlight the limitations of large, single-outlet MH tanks: low thermal conductivity, gas-phase transport resistance, and uneven pressure distribution hinder desorption in specific areas, thereby decreasing overall system performance.

The core temperature in the smaller tanks (D1) was monitored, as shown in Figs. 4(a) and 4(b). The cooling rate is similar to that at position B3 (near the wall) of the large tank and slower than that at any other location. This indicates that the interior of the small tank achieved more efficient heat compensation compared to the large tank. Notably, the hydrogen desorption flow rates are 1000 and 1125 sccm for the large and small tank configurations, respectively. According to Eq. 8, the smaller tank absorbs more reaction enthalpy per unit of time. This indicates a slower and more uniform decrease in temperature for the smaller tank, despite the elevated thermal load, and confirms the uniformity of the temperature distribution in both the axial and transverse directions of the tank. These findings support the split-tank strategy: the reduced thermal and hydrogen conduction paths, along with the increased surface area, enable the smaller tanks to improve the system's hydrogen desorption capability.

Fig. 4(c) illustrates the cumulative volume of released hydrogen in the peak flow period and the hydrogen flow rate. In both cases, the flow remains stable initially and then declines as the internal temperature drops. The large tank sustains its peak flow for a longer period, owing to its lower initial flow rate and enthalpy absorption per unit time. Further insights can be gained by considering the total volume of hydrogen released, as the small tank achieves a slightly better cumulative output. Based on hydrogen flow rates, the heat absorption of the MH is estimated at 22.9 W for the large tank (1000 sccm) and 25.1 W for the small tank (1125 sccm) using Eq. 8. This shows the greater hydrogen release capacity of the small tank despite a higher thermal load, confirming the efficiency of the modular design in managing transient heat transfer. During the decline stage, the small tank exhibits a slower decrease in flow rate, further indicating its superior heat exchange under natural convection. Additionally, its narrow interstitial spaces and staggered arrangement are expected to improve convective performance under forced or turbulent conditions by causing localized disturbances.

In summary, conventional large MH tanks suffer from two significant limitations: (i) low thermal conductivity causes severe overcooling at the core and local cessation of desorption, and (ii) gas transport resistance and uneven pressure distribution intensify temperature gradients, reducing overall hydrogen release. In contrast, smaller modular tanks alleviate these issues, and their performance is expected to be further enhanced by elevated temperatures and intensified airflow, resulting in superior desorption compared to large-tank designs.

3.2. Impact of exhaust heat recovery on PEMFC performance

The PEMFC was actively cooled using integrated fans, making its exhaust flow crucial for dissipating excess heat from the stacks. In terms of exhaust heat recovery, implementing a duct system to direct exhaust toward the MH tanks introduces potential concerns regarding thermal accumulation around the stacks, which may reduce the cooling efficiency [67]. Consequently, it is imperative to note that the proposed design does not interfere with the normal operations of the PEMFC. Current-voltage (I-V) measurements were conducted under two designs to evaluate the PEMFC performance: standard operation, and operation via the honeycomb configuration exhaust heat recovery system. As illustrated in Fig. 5(a), the I-V characteristics and corresponding power output curves under both conditions display almost identical behaviors over the current range. This consistency confirms that the honeycomb configuration system does not jeopardize the electrical stability or output performance of the PEMFC. The maximum output exceeds 300 W, meeting the power requirements of the H-bike.

In addition to evaluating the electrical performance, the thermal characteristics of the exhaust at various power levels were assessed. As depicted in Fig. 5(b), exhaust gas temperatures were recorded at three representative PEMFC outputs—0, 54, and 215 W—respectively corresponding to human-powered operation, flat-road cruising, and climbing a 30° incline on an electric-assist bicycle [57]. The measurements were used to assess the potential of exhaust heat as a passive heat source for warming the MH tanks. The exhaust temperature increases with the output power, reaching approximately 35 °C at an output of 215 W. This temperature range creates a favorable thermal environment conducive to hydrogen desorption. Even under partial loads, the exhaust temperature consistently exceeds ambient levels, indicating that sound thermal energy can be recovered across a broad spectrum of operating conditions.

3.3. Effect of PEMFC exhaust heat on hydrogen desorption

The effect of PEMFC exhaust heat on hydrogen desorption was examined under four conditions: natural convection and PEMFC power outputs of 0 W, 54 W, and 215 W. In the natural convection case, the MH tank was exposed to still ambient air at room temperature without fan-assisted airflow, so heat removal depended solely on natural circulation. By contrast, the 0 W case also lacked PEMFC heat input but included fan operation, creating forced convection around the tank. All tests used a single-layer tank with an initial hydrogen flow rate of 1000 sccm.

Fig. 6(a) illustrates the temporal evolution of the temperature at the center of the tank under various conditions. If external heat is insufficient, energy is drawn from the MH alloy and tank materials, causing a rapid drop in temperature. Across all four conditions, the initial desorption stage exhibits a nearly consistent cooling rate, as the PEMFC

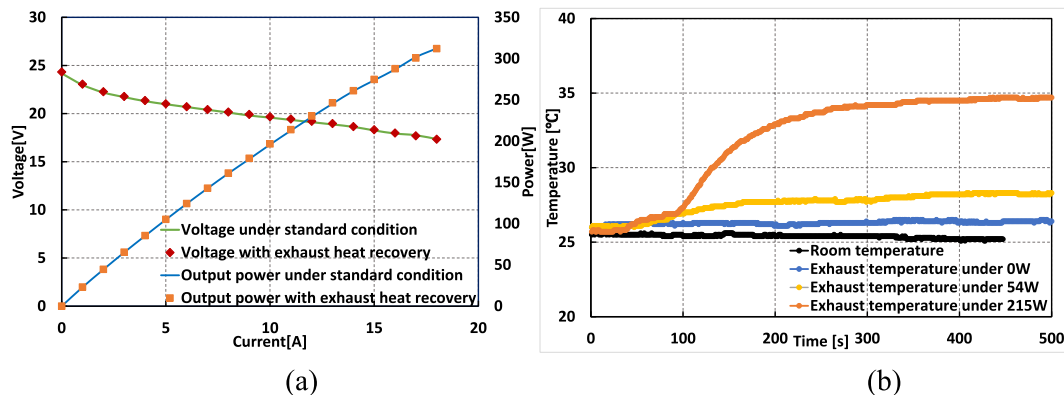


Fig. 5. Evaluation of PEMFC performance: (a) I-V characteristics under standard conditions using a honeycomb-structure design, and (b) exhaust gas temperatures at different power outputs.

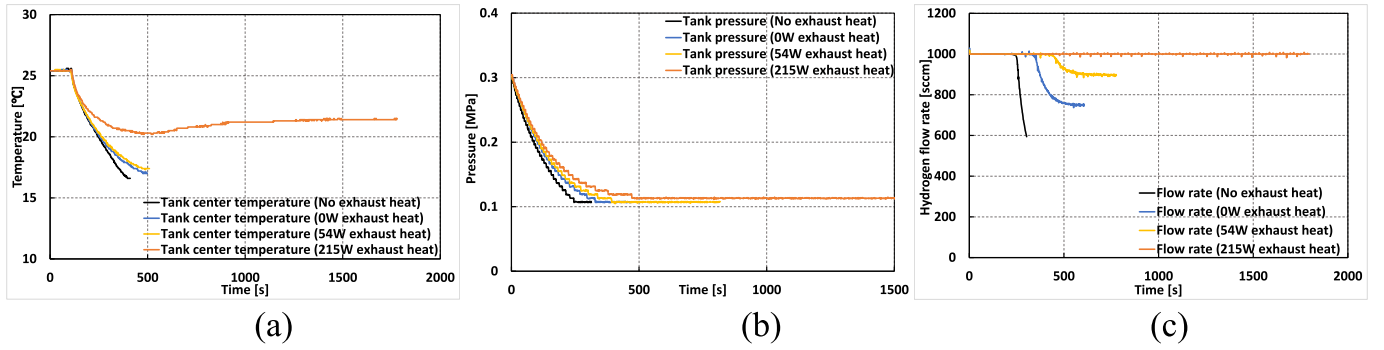


Fig. 6. Hydrogen desorption under different PEMFC exhaust heat levels: (a) temperature variation at the tank center, (b) internal pressure changes, and (c) hydrogen flow rate.

exhaust temperature shows little variation at the start and has a minimal impact on the tank center. As desorption progresses, the cooling rates under the four conditions diverge, reflecting differences in external heat input. The 215 W case performs best in slowing the temperature decline; however, even with exhaust heating, the temperature continues to fall, indicating that the absorbed heat remains below the approximately 22.9 W required to maintain a steady 1000 sccm flow. As the temperature continues to drop, the hydrogen flow cannot be maintained at 1000 sccm under natural convection or at 0 and 54 W. This is primarily due to the pressure-composition-temperature (PCT) characteristics of the MH, where a decrease in temperature sharply lowers the equilibrium pressure and desorption rate [68]. Internal temperature gradients exacerbate this effect, as areas that cool below the desorption threshold cease releasing hydrogen, thereby reducing the overall desorption capacity. Around 500 s under the 215 W condition, the temperature begins to recover due to limited internal heat transport and the increasing gradient between the surface and core. The low thermal conductivity of the MH alloy delays the core response, while the rising surface temperature enhances conduction inward. Over time, heat accumulates until a steady state is reached, where heat transfer to the tank balances the enthalpy absorption rate of 22.9 W, thereby stabilizing the temperature.

Fig. 6(b) illustrates the corresponding pressure variations within the MH tanks, with the internal pressure declining the fastest in the absence of exhaust heat. As the PEMFC output increases, the rate of pressure decline decreases, with the slowest decline observed at a power output of 215 W. The internal tank pressure is primarily governed by the dynamic equilibrium between the hydrogen released from the MH alloy and the hydrogen discharged from the tank outlet. Consequently, at a power output of 215 W, the slower rate of pressure decrease suggests that the system exhibits the highest hydrogen release rate from the MH alloy among the tested conditions. This occurs because as the temperature decreases, the equilibrium pressure also diminishes due to the PCT characteristics, reducing the thermodynamic driving force for hydrogen desorption. The continuous pressure decrease indicates that the rate of hydrogen desorption from the MH alloy remains below the target outflow rate of 1000 sccm. This imbalance ultimately results in a sharp decline in the hydrogen flow rate under the conditions of natural convection and power outputs of 0 and 54 W. In contrast, at a power output of 215 W, the higher thermal input not only helps maintain a higher average tank temperature but also increases the equilibrium pressure. When the internal pressure reaches approximately 0.12 MPa, the rate of hydrogen desorption from the MH alloy equals the hydrogen outflow rate of 1000 sccm, leading to a dynamic equilibrium. These findings confirm that 215 W of exhaust heat effectively supports stable pressure dynamics and continuous hydrogen desorption, further validating the superior thermal management capability of the system under this condition.

Fig. 6(c) illustrates the impact of the exhaust heat on the hydrogen

flow rate. In the absence of exhaust heat, the flow rate experiences a rapid decline, dropping below 1000 sccm within 300 s. At power outputs of 0 and 54 W, the decline starts at approximately 350 and 450 s, respectively. Conversely, the MH tank exhibits a stable flow rate of 1000 sccm for over 30 min at a power output of 215 W; the flow remains stable until the experiment is manually stopped. Considering the above analysis, the system is deemed to have reached a state of thermal equilibrium at this stage, where the enthalpy required for hydrogen desorption is effectively balanced by the heat absorbed from the external environment. Thus, most of the hydrogen stored in the tank can be efficiently released under this condition.

Table 2 presents a summary of the exhaust heat generated by the PEMFC, along with the enthalpy absorbed by the MH, under various conditions. The analysis is based on the period over which the hydrogen release rate was maintained at 1000 sccm. Under the output conditions of 0 and 54 W, the PEMFC exhausts 1.01 and 7.59 kJ of heat, respectively. On the other hand, the enthalpy absorbed by the MH is disproportionately high compared to the supplied heat, reaching 7.90 and 10.08 kJ, respectively. This indicates that the thermal energy provided under these circumstances is inadequate to satisfy the continuous enthalpy demand of the desorption reaction. In contrast, under a power output of 215 W, the total thermal energy supplied is 182.24 kJ, 41.22 kJ of which is absorbed by the hydride material. As the system reaches a state of thermal equilibrium, its heat utilization rate reaches 22.62 %. As such an equilibrium state was not achieved under the other conditions, the utilization rates could not be accurately determined in those cases; however, the values are lower than 22.62 %.

In conclusion, the condition involving a power output of 215 W surpasses all other conditions in maintaining elevated internal temperatures, a slower pressure decay rate, and a sustained hydrogen flow rate, leading to enhanced thermal and desorption stability. It confirms that effectively recovering PEMFC exhaust heat significantly enhances the thermal stability of MH tanks, mitigates internal pressure decay, and increases the period of stable hydrogen supply. The proposed integration strategy promotes effective thermal coupling, eliminating the need for external heating and enhancing the overall system performance and energy efficiency.

3.4. Effect of the layers of cartridge on the hydrogen supply

The effect of the number of MH cartridge layers on the system's

Table 2

Exhaust heat and heat absorption under different PEMFC exhaust heat conditions.

Exhaust heat level	No heat	0 W	54 W	215 W
Exhaust energy (kJ)	–	1.01	7.59	182.24
Absorbed heat (kJ)	5.73	7.90	10.08	41.22
Exhaust heat recovery efficiency (%)	–	–	–	22.62

capacity to maintain the designated hydrogen flow rate was evaluated experimentally under a PEMFC power output of 215 W. The flow rate was initially calibrated at 2000 sccm, corresponding to the estimated maximum hydrogen demand for H-bike operation on an inclined terrain [57]. The duration of maintaining this flow rate was regarded as the primary indicator of desorption performance and a measure of the system's ability to support continuous H-bike operation during peak power demand. In this context, 215 W represents the most extreme operating condition, and demonstrating stable performance under this scenario indicates that hydrogen demand can be met across the full range of practical applications.

As illustrated in Fig. 7, the exhaust temperature exceeds 35 °C under a power output of 215 W. In this case, while the single-layer tanks fail to sustain the target flow rate of 2000 sccm in the first 250 s, the two-layer tanks maintain a stable hydrogen flow rate of 2000 sccm throughout the entire 30-min testing period. This illustrates that single-layer tanks are inadequate for sustaining prolonged periods of operation. In contrast, configurations incorporating two layers can maintain the desired hydrogen flow rate over an extended period. The ability to consistently deliver 2000 sccm of hydrogen for 30 min further signifies that the system achieves a state of thermal equilibrium, where the heat absorbed from the PEMFC exhaust is sufficient to continuously facilitate endothermic desorption. Based on this thermally balanced state, the corresponding thermal input and hydrogen desorption power were estimated. At a power output of 215 W, the effective thermal output at a steady exhaust temperature is approximately 114.81 W. In the two-layer configuration, operating at 2000 sccm, the enthalpy absorption power reaches 45.8 W, corresponding to a utilization efficiency of 39.90 %. Similarly, in the single-layer case, with a flow rate of 1000 sccm, the absorption power was estimated at 22.9 W, yielding a utilization rate of 19.95 %. These results confirm that with sufficient exhaust power, increasing the number of MH layers significantly enhances heat utilization. The two-layer configuration is sufficient to meet the maximum hydrogen demand, ensuring the complete release of hydrogen stored within the cartridges. Beyond this two-layer setup, the PEMFC stack possesses a baseline weight of 5 kg and a volume of 6.1 L, with each additional MH layer contributing approximately 1.36 kg and 1.26 L. Augmenting the number of layers also increases the thermal distance between the exhaust source and the outer cartridges, which may diminish heat transfer efficiency and induce pressure drops. Nevertheless, as demonstrated by Eq. 8, allocating the thermal load across a greater number of cartridges enables each layer to attain higher steady-state temperatures, thus enhancing overall heat utilization efficiency. The superior performance of multilayer configurations is attributable to the semi-independent operation of each cartridge and the increased heat exchange surface area provided by the honeycomb structure, which extends the exhaust-tank interaction period and optimizes heat recovery.

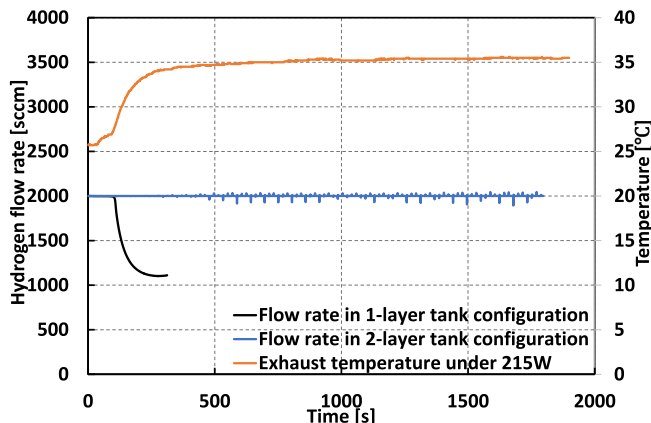


Fig. 7. Effect of number of MH tank layers on hydrogen flow rate.

Table 3 compares the current system with previous methods of waste heat recovery from FCs. The results show that conventional proton exchange membrane fuel cell-metal hydride tank-heat exchanger (PEMFC-MHT-HX) systems can achieve thermal recovery efficiencies of up to 36.8 % with moderate control complexity at large scale; while a large system of open-cathode PEMFC combined with multiple MH canisters can recover 32–40 % with lower complexity but are mainly suited for stationary applications. Solid oxide fuel cell-organic Rankine cycle (SOFC-ORC) hybrids provide less than 30 % recovery and involve high system complexity. Proton exchange membrane fuel cell-thermoelectric generator (PEMFC-TEG) systems recover less than 1 % of the energy due to the low temperature gradients. In contrast, the proposed honeycomb multilayer PEMFC-MH setup achieves a utilization efficiency of 39.90 % under 215 W exhaust conditions, featuring a simple design optimized for small-scale H-bike applications. This demonstrates the benefits of modular multilayer MH integration for better waste heat utilization in portable fuel cell systems.

4. Conclusion

This study presents a novel tank structure and honeycomb configuration, offering an integrated hydrogen supply system that utilizes exhaust heat from PEMFC to enhance desorption performance in MH tanks. The key findings are summarized as follows:

1. The split-tank strategy shows that dividing a large MH tank into smaller cartridges effectively reduces heat conduction and hydrogen diffusion paths, enhances thermal uniformity, and relieves localized overcooling. This greatly improves hydrogen release capacity by addressing the thermal bottlenecks present in large tanks.
2. Effective heat recovery with a honeycomb tank setup. The honeycomb tank passively captures exhaust heat without impacting the PEMFC's cooling efficiency or power output. At a power level of 215 W, the system reaches thermal equilibrium and maintains a steady hydrogen flow rate of 1000 sccm for over 30 min. The heat recovery efficiency of 22.62 % demonstrates the viability of direct thermal coupling between the PEMFC and MH.
3. Using multilayer cartridges increases the effective heat exchange area and distributes the thermal load, allowing each layer to reach higher steady-state temperatures. With two layers, the system sustains a hydrogen flow of 2000 sccm for 30 min at 215 W, achieving a utilization efficiency of 39.90 %. This confirms both scalability and improved thermal utilization efficiency compared to single-layer designs.

Overall, these strategies tackle the main thermal challenges of MH tanks—low thermal conductivity, uneven pressure distribution, and limited heat recovery. By improving heat transfer, boosting desorption stability, and increasing heat utilization efficiency, the proposed design offers a strong path for integrating PEMFC exhaust heat into MH systems. This modular and scalable approach provides a practical solution

Table 3
Comparative analysis of waste heat recovery strategies.

Ref.	Structural configuration	Thermal recovery	Control complexity	Scale
Wang et al. [52]	PEMFC-MH-HX	29 %–36.8 %	Medium	Large
Omrani et al. [54]	Large scale PEMFC-MH	32 %–40 %	Low	Large
Mouissi et al. [69]	SOFC-ORC	<30 %	High	Large
Zhang et al. [70]	PEMFC-TEG	<1 %	Medium	Small
Present Work	Honeycomb multilayer PEMFC-MH	39.90 %	Low	Small

for lightweight hydrogen mobility and other small-scale fuel cell applications.

5. Future Work

Although thermal equilibrium was successfully achieved using an output power of 215 W, allowing for the release of most of the stored hydrogen, the internal tank pressure stabilized at approximately 0.12 MPa. Given the diverse pressure requirements of commercial PEMFC systems, future research efforts will focus on maintaining elevated minimum tank pressures by increasing the number of MH tank layers beyond that of the existing configuration. These considerations further emphasize the need for a five-layer design to ensure adequate hydrogen release and compatibility with a broader range of PEMFC systems.

CRediT authorship contribution statement

Shan Miao: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation. **Tomoya Ezawa:** Investigation. **Masami Sumita:** Investigation. **Koki Harano:** Investigation. **Ayane Imai:** Investigation. **Noboru Katayama:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Kiyoshi Dowaki:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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